

Hsi-Che Lin

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Education

National Taiwan University

MS IN GRADUATE INSTITUTE OF COMMUNICATION ENGINEERING

- Advisor: Yu-Chiang Frank Wang
- Research Topics: Multi-modal large language models, Efficient AI

Taipei, Taiwan

Sep 2024. - present

National Taiwan University

BS IN ELECTRICAL ENGINEERING (GPA: 4.17/4.30, RANK: 19/202)

Taipei, Taiwan

Sep. 2021 - Jun. 2024

Publications

[1] EMLoC: Emulator-based Memory-efficient Fine-tuning with LoRA Correction [Under review]

Hsi-Che Lin, Yu-Chu Yu, Kai-Po Chang, Yu-Chiang Frank Wang

- Developed a novel fine-tuning method reducing memory usage by a low-rank compressed alternative of the model.
- Enable fine-tuning **38B VLM** and **12B Diffusion Model** with just **24GB VRAM**, outperforming existing baselines.

[2] Improving Speech Emotion Recognition in Under-Resourced Languages via Speech-to-Speech Translation with Bootstrapping Data Selection [ICASSP'25]

Hsi-Che Lin*, Yi-Cheng Lin*, Huang-Cheng Chou, Hung-yi Lee

- Propose a pipeline using **synthesized data** and **bootstrapping selection** to improve under-resourced SER.
- Outperform other augmentation methods and demonstrate generalizability across diverse models and languages.

[3] QuAVF: Quality-aware Audio-Visual Fusion for Ego4D Talking to Me Challenge [CVPR'23 Workshop]

Hsi-Che Lin, Chien-Yi Wang, Min-Hung Chen, Szu-Wei Fu, Yu-Chiang Frank Wang

- Designed a **multi-modal framework (vision, audio, and facial landmarks)** to achieve robust detection results.
- Secured 1st place on the leaderboard, achieving state-of-the-art performance.

[4] Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks [ICLR'25]

Chien-yu Huang, Wei-Chih Chen, ..., **Hsi-Che Lin**, ..., Hung-yi Lee

- Introduce several evaluation tasks beyond classification, including regression and sequence generation tasks.
- The largest and most comprehensive open-source benchmark for spoken language models.

[5] ReXTime: A Benchmark Suite for Reasoning-Across-Time in Videos [NeurIPS'24 D&B]

Jr-Jen Chen, Yu-Chien Liao, **Hsi-Che Lin**, Yu-Chu Yu, Yen-Chun Chen, Yu-Chiang Frank Wang

- Develop an automated pipeline for generating a large-scale temporal reasoning question-answer data.
- The first benchmark for video understanding focusing on **cause and effect reasoning**.

[6] On the social bias of speech self-supervised models [INTERSPEECH'24]

Yi-Cheng Lin, Tzu-Quan Lin, **Hsi-Che Lin**, Andy T. Liu, Hung-yi Lee

- Conduct a comprehensive analysis of various factors that influence social bias and methods to debias.

Experience

Deep Learning for Computer Vision (120 students), Teaching assistant

Fall 2023-2025

- Lead a team of 12 TAs (2025), Design homework about VLM (2024), SSL and segmentation (2023)

Machine Learning (650 tudents), Teaching assistant

Spring 2025

- Design homework assignment about diffusion model customization

Introduction to Generative AI (1000 students), Teaching assistant

Spring 2024

- Design homework assignment about LLM fine-tuning

Projects

E.SUN AI Open Competition 2021 Winter-Credit Card Product Category Prediction

Top 18% of 859 Teams

- LSTM model ensemble for time-series data with feature engineering to reduce the input dimension and complexity.

Kaggle "G-Research Crypto Forecasting" Challenge

Top 9% of 1946 Teams

- Developed a redictive model (Light Gradient Boosting Machine) with feature selection.

Skills

Programming languages: Python

- Libraries: PyTorch, transformers, diffusers
- Models & Architectures: CLIP, InternVL, Llama, Diffusion models